

Lung Nodule Segmentation — DynUNet / 3D U-Net — sliding-window (128^3)

1. Task

Voxel-level segmentation of pulmonary nodules in 3D chest CT volumes, lung-bbox-cropped. Same task as the paper's `dynunet_v2` model — the only difference is inference recipe: **sliding-window over 128^3 patches at native resolution** rather than a single forward on a 256^3 resample.

- **Input** (1, H, W, D) CT crop, intensity-normalised to [0, 1], lung-bbox + 20-voxel padding, kept at native voxel grid.
- **Training input** 128^3 patches sampled via `RandCropByPosNegLabeld` at a ratio of `pos:neg = 2:1`, `num_samples = 4` per volume.
- **Inference** MONAI `sliding_window_inference`, ROI (128, 128, 128), `overlap = 0.5`, `sw_batch_size = 4`, Gaussian blending, single output at the input resolution.
- **Output** (2, H, W, D) softmax logits: 0 = background, 1 = nodule.

2. Model

Property	Value
Architecture family	DynUNet / 3D U-Net
Spatial dims	3
Input channels	1
Output channels	2 (softmax)
6-level auto-configured	strides=[1,2,2,2,2], kernel=3
Trainable parameters	16,538,530

3. Data

Same unified corpus + split as the paper's 256^3 baselines:

Source	Role
NLST	train + val + test
NSCLC-Radiomics	train + val + test
LIDC-IDRI	train + val + test

- **Split:** `unified.json` — patient-grouped, dataset-stratified. 1 683 train / 297 val / 325 test series.

- **Lung crop:** per-series 3D bbox from `bboxes_unified.json` (lung ROI + medium-model fallback), + 20-vox padding. No resampling.
- **Class imbalance:** nodule voxels are $\sim 10^{-5}$ of the total — see §7.

4. Training

Setting	Value
Loss	Focal Tversky + weighted CE ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$, $\lambda_{CE}=0.1 / 0.3$, nodule class weight = 100)
Optimizer	Adam
Learning rate	1×10^{-5}
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{max} = 400$, $\eta_{min} = 1 \times 10^{-6}$
Batch size	2 volumes $\times$ 4 patches = 8 patches / step
Patch size	128^3 (positive-biased random crop, <code>pos:neg = 2:1</code>)
Epochs	400
Random seed	42
Mixed precision	bf16 (autocast, no GradScaler)
Augmentation	3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast
Hardware	1 $\times$ NVIDIA H100 80 GB (fat01)

Best checkpoint recorded at **epoch 264 / 400** (see `_meta` in metrics JSON).

5. Evaluation protocol

Evaluated on the `unified` **test** split (325 series held out — never seen during training or model selection). Predictions taken as `argmax` over the 2-channel softmax output from `sliding_window_inference`. All voxel-level metrics are micro-averaged over the whole split.

6. Results

Ticket-required metrics

Metric	Value
mean IoU	0.6557
Accuracy	0.9985

Metric	Value
Precision	0.3379
Recall	0.8086

Supplementary

Metric	Value
IoU (foreground / nodule class)	0.3129
IoU (background class)	0.9985
Dice / F1 (micro)	0.4767

Per-case Dice distribution (325 test cases)

Statistic	Value
Mean	0.4219
Std	0.2577
Min	0.0000
p05	0.0047
p25	0.2078
Median	0.4386
p75	0.6311
p95	0.8233
Max	0.9219

Voxels evaluated: 7,849,317,259. Cases: 325.

7. Notes on interpretation

Sliding-window inference at 128^3 trades global context (a 256^3 resample sees the whole lung in one pass) for preserved resolution (no down-sampling of small nodules). On this unified corpus the trade tends to be neutral-to-negative on nodule Dice — small-nodule gains do not compensate for lost global context on typical test cases.

Class-imbalance caveats are unchanged from the 256^3 reports: Accuracy is trivially ~ 1 , mIoU is dominated by IoU_bg, and Dice / F1 (micro) is the honest voxel-level summary. Per-case mean weights each patient equally regardless of nodule volume.

8. Reproducibility

- Config `/home/werner/nodule-training-demo/configs/dynunet_sw.yaml`
- Checkpoint `stage2_fine/checkpoints_dynunet_sw/best_model.pth`

- Metrics JSON `/tmp/claude-1002/-home-werner-nodule-segmentation/c2faa0a0-5a75-48f5-8288-377df40092b7/scratchpad/sw_metrics/dynunet_sw.json`
- Command `python stage2_fine/eval_metrics.py --config /home/werner/nodule-training-demo/configs/dynunet_sw.yaml --checkpoint stage2_fine/checkpoints_dynunet_sw/best_model.pth --split test`